



A metamodel of an informational structure for model-based technology roadmapping

Ilya Yuskevich^{a,b,*}, Andreas Makoto Hein, PhD^a, Kahina Amokrane-Ferka, PhD^a, Abdelkrim Doufene, PhD^a, Marija Jankovic, PhD^b

^a IRT SystemX, 8 Avenue de la Vauve, 91120 Palaiseau, France

^b Université Paris-Saclay, CentraleSupélec, Laboratoire Génie Industriel, 3 Rue Joliot Curie, 91190, Gif-sur-Yvette, France

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ABSTRACT

Recent contributions in the field of technology roadmapping often aim to apply various numerical models and tools to facilitate the roadmapping process and enrich its outcomes. This trend resulted in the emergence of so-called model-based technology roadmapping. We consider it as the future development of the traditional document-based paradigm. One of the general approaches to support the model-based roadmapping is to develop a roadmap's metamodel that would define it independently from the application context and link it to the existing roadmapping literature. In this paper, we attempt to create such a metamodel by generalizing and formalizing existing document-based roadmaps. We validate our metamodel via reproducing three very different roadmaps from the literature, not included in the set of roadmaps from which the metamodel was created, using the novel formal approach. The fact that these roadmaps were reproduced using the proposed metamodel indicates its applicability to many classes of real roadmaps. The results of this work seem beneficial for architects of the software tools for roadmapping and for the regular participants of roadmapping sessions.

1. Introduction

Large enterprises, such as Lockheed, NASA, Motorola, Philips – just to name a few – have been successfully practicing technology roadmapping (TRM) over the previous 40 years (Kerr and Phaál, 2020). Domain experts consider TRM as an efficient technique for long-term planning of technologies, innovations, programs, products, and services.

TRM is a process that produces a technology roadmap. Phaál et al. (2004) define a technology roadmap as a "time-based chart, comprising a number of layers that typically include both commercial and technological perspectives." In practice, roadmaps serve the internal and external communication of plans, decisions, and responsibilities and help to focus on long-term goals of high-priority (Albright and Kappel, 2003).

Originating from large technological enterprises' practices, TRM now encompasses various applications and organizational contexts. Phaál et al. (2004) distinguish eight classes of roadmaps by the object of planning, such as products, capabilities, strategies, long-range goals, knowledge assets, programs, processes, and operations of complex system integration. Kim et al. (2018) introduce design roadmapping and

provide an example of roadmapping applied to a single-product user-experience-focused startup.

A traditional roadmapping process is typically done through a series of workshops and involves a collaboration of multiple experts from different domains. This process is informationally- and labor-intensive, and leads to several known difficulties. Following the digitization trend, researchers have proposed numerous approaches to support TRM with computer-based tools. Most of the propositions came from the realm of computational linguistics and related to the automated analysis of large textual databases, such as research papers and patents (Kostoff and Schaller, 2001).

Knoll et al. (2018) consider the roadmapping problem from a systems engineering perspective. Indeed, a large number of roadmap elements and relationships between them provide grounds to perceive a roadmap as a system. Following the model-based paradigm in engineering design, Knoll et al. (2018) have proposed the notion of model-based technology roadmapping (MB-TRM). The distinct feature of this MB-TRM is that it uses executable conceptual design models as building blocks of a roadmap. The fact that these models are stored and processed in a computer helps to create, maintain, and visualize the roadmap. Besides, it allows

* Corresponding author.

E-mail address: ilya.yuskevich@gmail.com (I. Yuskevich).

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us to define data processing functions within the scope of a digital roadmap that would perform operations useful for decision-makers (e.g., summarizing, verifying, assessing, extrapolating). This MB-TRM was initially proposed for a specific organizational context – a large international corporation that produces complex technical systems and systems-of-systems (Knoll et al., 2018), and no general model for MB-TRM seems to exist. Such a generalized MB-TRM could be applied to all types of contexts – from a startup to a space agency.

This paper proposes such a general model for MB-TRM, which can be used to model domain-specific roadmaps. We call such a general model a metamodel. According to Mellor (2004): "A metamodel is a model of a modeling language. It defines the structure, semantics, and constraints for a family of models." A model, according to Mellor (2004), is "a simplification of something <..> It consists of sets of elements that describe some physical, abstract, or hypothetical reality."

Generally, the authors argue for metamodels via the following arguments (Ober and Prinz, 2006; Terrasse et al., 2006):

- Metamodels give "an inspiration", a starting point for the developers of domain-specific models;
- With certain reservations, metamodels may serve as an integrative basis for diverse models, fully or partially built upon the same metamodel;
- Metamodels also "lift up the abstraction level", which allows theoreticians and practitioners discuss essential concepts of some theory ignoring specificities;
- Description of the complex domain is challenging due to not coherent terminology, may require syntax of various languages, and often composed of "partial descriptions". In such cases metamodels are useful as compact representations of the broad body of knowledge.

Therefore, the ultimate goal of the proposed metamodel is to attain the aforementioned advantages in the application to roadmapping field. More specifically, the development of the metamodel is an important contribution to the model-based roadmapping domain that, as it was shown in previous papers (Golkar and Garzaniti, 2020; Gradini et al., 2019; Knoll et al., 2018), has the potential to considerably facilitate TRM process.

In view of MB-TRM development, we identify the following research questions:

RQ1. Can a metamodel be proposed that is compatible with existing definitions of a technology roadmap?

RQ2. Can the instantiability of the proposed metamodel be demonstrated in various contexts?

RQ3. Is the proposed metamodel aligned with principles of MB-TRM?

RQ4. Is the proposed metamodel usable? To approach usability it can be, for instance, assured that metamodel is internally consistent, concise, and built following best practices.

The rest of the paper is structured as follows. In section two, we discuss the related literature. In section three, we describe our research methodology – the processes of metamodel construction and validation. The results are presented in section four. The final sections are for discussion, future work, and conclusions.

2. Background

In the following, we first review current trends in the technology roadmapping domain. Second, we provide a review of the literature on TRM assisted with digital tools. Then, we discuss the difficulties of current TRM practice to support the arguments for MB-TRM. Finally, we summarize the actual papers on MB-TRM and highlight the defining principles of the approach we present in the following sections.

2.1. Research streams in technology roadmapping domain

De Alcantara and Martens (2019) conducted a bibliometric analysis of current TRM practice and built citation clusters. Based on this review and on our own observations, we would highlight three major research streams in this field.

First, many research groups working on the improvement of generic TRM process by combining it with other technology and innovation management frameworks. Cheng et al. (2016); Pagani (2009); Strauss and Radnor (2004) worked on integration of TRM with scenario planning, an important approach when it comes to the forecasting of the ambiguous reality. Integration with various business modeling frameworks helps to foster innovative and sustainable companies (Abe et al., 2009; Despeisse et al., 2017; Schaller et al., 2018). View on TRM through business ecosystem perspective and its tailoring to the needs of small and middle enterprises presented in Battistella et al., (2015). Al-Ali and Phaal (2019) incorporated agile principles into TRM in order to help organizations achieve digital transformation.

Second stream is the integration of the roadmapping with various decision-making techniques. Aleina et al. (2017) complemented TRM with various optimization and ranking approaches in order to perform space exploration technologies prioritization based on multiple criteria. Bloem da Silveira Junior et al. (2018) presented approach integrating Delphi, morphological analysis, decision matrix, and prioritization. Another fairly popular approaches that complement TRMs are analytical hierarchy process (Geum et al., 2014; Jeon et al., 2011) and quality function deployment (An et al., 2008; Lee et al., 2013; Noh et al., 2021).

The third stream relates to the digital tools and automation of the roadmapping procedures, which is of particular interest to the current contribution and will be reviewed in the next section.

2.2. Computer-aided roadmapping

Kostoff and Schaller (2001) were apparently the first who classified the roadmapping process into expert-based, computer-based, and hybrid. They define the expert-based roadmapping as an approach in which "a team(s) of experts is convened to identify and develop attributes for the nodes and links of the roadmap," and computer-based as an approach where "large textual databases that describe science, technology, engineering, and end products are subject to computer analyses." Kostoff and Schaller (2001) acknowledge that a fully automated process of roadmapping is not feasible at the moment. They, therefore, suggest a hybrid procedure where computer assistance complements the process of expert negotiation.

Zhang et al. (2016) contribute to the hybrid TRM by proposing a TRM framework for technical intelligence, including text mining, bibliometrics, and several qualitative expert-based approaches. The interesting takeaway in the context of our work is that Zhang et al. (2016) designed a structure of a hybrid roadmap. This structure, or, we may say, syntax consists of labels L , implications I , time T , objects $O(L, I, T)$, and the relationships between objects $R(O1, O2)$. It serves then not only as a syntactic pattern for text mining but also as a description language for a roadmap.

Other examples of TRM assisted by text-mining could be found in Kajikawa et al. (2010) and Miao et al. (2020). Kajikawa et al. (2010) propose to build citation networks to find an ontology of emergent technologies. The results of this technology forecasting procedure are used in the following phase of an expert-based roadmap development. Miao et al. (2020) utilize Technology-Relationship-Technology syntactic pattern to build innovation pathways and, consequently, a roadmap. The aforementioned papers exploit computer-based tools mainly for data gathering and preprocessing. Our contribution differs from them by aiming to propose a formal architecture of a roadmap itself. Such formal structure would help to partially automate and instrumentalize the work of experts engaged in the real-time process of roadmap development.

2.3. Document-based TRM difficulties and foundations of MB-TRM

Kim et al. (2018, 2016) identified several challenges in current roadmapping practices through a series of semi-structured interviews:

- Even though one of the main functions of roadmapping is the communication of plans, there is still a conflict, especially when the communication happens between multidisciplinary teams;
- TRM lacks a feedback loop from final users of a product, so the focus on creating the value for a customer is not followed;
- Lack of flexibility of a roadmap in the fast-changing environment; necessity to keep roadmap 'alive' leads to a request of an agile and iterative process;
- Inability to predict the future, hence often the "plan is not followed."

McMillan (2003) brought up several lessons learned from his practical experience in Rockwell Automation:

- Senior management must demonstrate a commitment to the process;
- Common definitions should be established for elements of a roadmap, but the process should be flexible enough to re-define these elements;
- Business units should at some point "endorse and own the process" of roadmapping;
- Some business units perceive roadmapping as a "mandated exercise to be completed on the command of management" because this process's benefits were not well understood.

Gradini et al. (2019) suggest that some of these difficulties may be resolved by a major shift from a document-based to a model-based technology roadmapping.

MB-TRM implies a formal architecture of a roadmap. Phaah and Muller (2009) laid the foundation for it. They divided a roadmap architecture into two realms: informational structure and graphical style, which corresponds to the famed Model-View-Controller pattern from Computer Science. The informational structure of a roadmap essentially consists of hierarchic elements (technologies, resources, components, products, markets, etc.) and relationships between them. Visual style (Kerr and Phaah, 2015) typically reflects an internal structure through a layered chart of a directed graph.

Gradini et al. (2019) and Knoll et al. (2018) present a different view in their MB-TRM framework. They define an architecture of separate roadmap elements (e.g., systems) that must be instantiated, parameterized, and linked to other elements (e.g., technologies) during collaborative design sessions. Linked elements thus turn to nodes of a centralized graph (i.e., a roadmap). Time and layers are implied but not manifested in the methodology as explicit dimensions.

Golkar and Merzaniti, (2020) promote further the concept of MB-TRM by introducing Model-Based Technology Roadmap Architecture. They argue that model-based roadmapping should consist of five key elements (functions): model structure, data analytics, knowledge management, visualization, and continuous improvement. The roadmap is built around particular class of systems (in their case study Orbital Launch Vehicles) and includes portfolio of products, services, technologies, and components, related to this system. The system of interest is modeled using Object Process Model (OPM) visual language, and is complemented with parametric models to calculate system's Figures of Merits.

Several contributions introduce processes that may be integrated into MB-TRM software as useful functions. Geum et al. (2014) incorporated system dynamics simulations to 'linked grid roadmap' to analyze different scenarios of a new service launch. Equations that establish relationships between external influencing factors and service key performances are examples of transfer functions. Lu and You (2018) proposed metrics to assess the importance of nodes (in their case, weapon systems) for technology infusion based on the topology of a

graph (an example of a graph traversal function). Yuskevich et al., (2018, 2021) proposed an algorithm to assess and forecast the technology rate of change based on historical data (an example of a trend extrapolation function).

The results of our literature review, first of all, highlight the problems of existing TRM practices, which provide the necessary motivation to further develop MB-TRM. Indeed, the main body of literature focuses on the enhancement of the TRM's value for stakeholders, tailoring the process to specific contexts, or on the refinement of the decision-making. Existing computer-based approaches aim at better data gathering with natural language processing. However, solving the problems associated with TRM in general has been rarely seen through the perspective of building potentially complex interactive informational systems (examples of such systems now emerging in the realm of software product management). At the same time, previous publications on MB-TRM considered domain-specific roadmaps. Therefore, this research aims to address this gap by proposing a domain-independent metamodel. The purpose of this metamodel is two-fold. In the theoretical realm, it contributes to a definition of a field by raising the level of abstraction. And from a practical point of view, it provides a starting point for developers of roadmapping software.

Based on the results of our review, we argue that there are two determining features of MB-TRM. The first is the formal syntax, which, along with the agreed semantics, determines elements of a roadmap and relationships between them. The second is a centralized database allowing for collaborative work and compatible with defined syntax and semantics. Availability of functions, different from basic create, read, update, and delete (CRUD), is beneficial but optional.

The latter differentiates MB-TRM also from what Kostoff and Schaller (2001) address as a computer-based approach, as the definition prescribes the presence of complex computer analysis. MB-TRM can be considered as a special case of hybrid TRM, with a mandatory requirement of formal syntax and defined semantics.

The fact that elements of a model-based roadmap are created, stored, and maintained numerically facilitates all processes and brings flexibility unachievable for document-based roadmapping. We would emphasize here that simple digitization of documents or transferring them to spreadsheets is not an objective. By documents, we mean both paper and digital artifacts.

3. Research methodology

To validate the proposed metamodel, we will use an ad-hoc procedure inspired by existing methodologies from Model-Driven Engineering (MDE) and Semantic Web domains. We will use a modern language workbench (a software for Domain-Specific Languages (DSLs) design). Our success criterion is to satisfy the objectives of the validation formulated in the following.

3.1. Objectives of validation

To address metamodel development and validation, we propose to adapt the approach of meta ontology O2 of Gangemi et al. (2006) (Figure 1). On this level, a metamodel expresses the intended conceptualization of some domain by rational agent(s). The intended conceptualization is influenced by pragmatic objectives and semantic space, which, generally speaking, may differ between rational agents. Typically, Metamodeling then becomes an exercise in finding a social consensus on the "right" metamodel.

Our objective here is not to find a metamodel for TRM based on social consensus between rational agents. Instead, our objective is to construct a sufficiently general metamodel for modeling a wide range of existing roadmaps.

More specifically, we will focus on the verification of the following statements:

S1. DSLs instantiated from our metamodel correctly express several

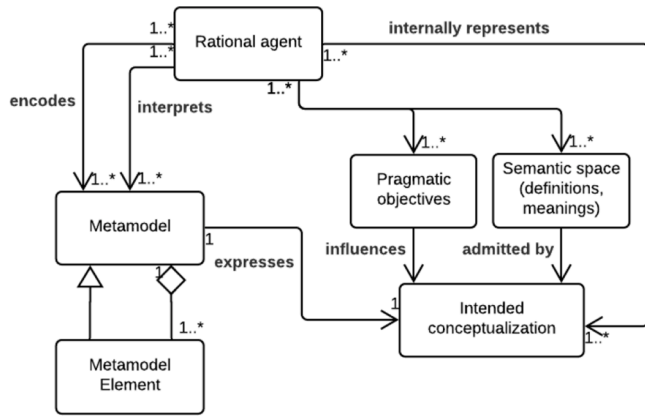


Fig. 1. Conceptual model of Metamodeling (originated in Gangemi et al. (2006))

publicly available technology roadmaps.

S2. The roadmaps that we selected for validation belong to significantly different contexts.

S3. Our metamodel can be used as a prototype of an MB-TRM software system, supporting CRUD-operations and graph-traversal functions.

If S1 holds, then our metamodel is internally consistent (RQ4) and compatible with definitions of a technology roadmap (RQ1). S2 asserts that our meta model constitutes a useful generalization for a practice of TRM (RQ2). S3 is a formalized version of RQ3.

3.2. Validating the Metamodel

To design our approach to metamodel validation correctly, we propose to review the MDE literature. MDE is a branch of computer science, promoting the further rise of the abstraction level in software, i.e., from general-purpose programming languages to platform-independent models.

We propose to base ourselves on the example-driven metamodel validation procedure from (López-Fernández et al., 2015)). López-Fernández et al. (2014) highlight two broad classes of validation objectives: "are we building the right metamodel?" and "are we building the metamodel right?" The former aims to validate the correspondence of metamodel concepts to definitions and meanings accepted in the domain of interest. The latter's goal is to verify that the metamodel is consistent and meets the standards and best practices of MDE, ensuring that a metamodel is reusable.

López-Fernández et al. (2015, 2014) proposed an example-driven procedure of metamodel construction and validation. They also developed a tool that enables semi-automatic metamodel construction from several model fragments created by domain experts. The resulting metamodel can then be validated by testing it against valid and invalid models.

Since metamodels are built for practical purposes, they can also be validated by direct instantiation for a specific use case. As an example, García-Holgado and García-Peñalvo (2017) performed such validation for learning ecosystems.

The process of metamodel construction and validation is shown in Fig. 2. The process starts with a review of published document-based technology roadmaps. During the review, a metamodel creator (first author of this article) identifies common elements among various documents, conceptualizes and classifies them. The first version of a metamodel (level M2 of MOF architecture) is a product of this step. In the second step, we select several roadmaps with significantly different structures and application domains. Then, for each of these roadmaps, the application domain-specific model (M1) is instantiated from a metamodel (M2). In the third step, each roadmap (user data M0) is

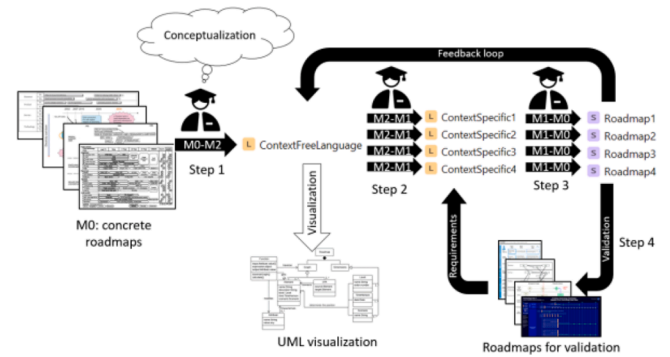


Fig. 2. The iterative process of metamodel construction and validation

instantiated from a corresponding application domain-specific model (M1). Produced data should be identical to the data contained in the selected documents. If we failed to go through all the steps, we can modify a metamodel and repeat the instantiation. The process needs to be repeated iteratively until a metamodel is mature enough.

The process of choosing roadmaps for this study prone to selection bias. There were no specific measures taken to mitigate this risk. This is one of the major limitations of our methodology. We picked published roadmaps based on the following criteria:

- To accurately conceptualize the domain of interest, the completeness of the information is an important consideration. However, in the most cases, the authors tend not to disclose the sensitive roadmap information fully, usually not going beyond schematic representation. Hence, the pool of available options is, in fact, quite limited;
- Generalizability is an essential property of any valid metamodel. We attempted to select roadmaps belonging to different industries, made in organizations of different sizes and positions, served to different purposes.

3.3. Software tool for the metamodel validation

To support the procedure, we selected a tool that assists in the creation and composition of DSLs (a language workbench). We have chosen JetBrains MPS – a modern language workbench that allows us to define an abstract and concrete syntax (structure and editor in MPS-terms) of a DSL and write code in a new language (create models) with projectional editing. The projectional editor allows direct manipulation of an abstract syntax tree (AST) of a model. The latter is important for metamodel validation – any expression incompatible with the syntax defined for DSL cannot modify an AST of a model. In this way, we automatically ensure the internal consistency of a metamodel.

[Note: for more information on MPS, visit ("Glossary - Help |MPS", 2020)].

Language composition is presented in Fig 3. Both the application domain-independent language (metamodel M2) and the application domain-specific language (model M1) are written in the Base Language – a Java-like language for DSL definition in MPS. The operation of metamodel instantiation is equivalent to language extension in JetBrains MPS. User data (in our case, a roadmap) is defined using the concrete

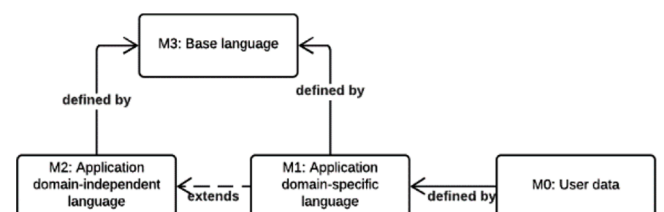


Fig. 3. Language composition structure

and abstract syntax of application domain-specific language (M1).

We chose a language workbench in which an AST is represented in textual form by default. For the sake of clarity, we will present an abstract syntax of the DSL in the form of a UML class diagram (for concepts) augmented with OCL expressions (for constraints); and MO roadmaps in a textual form aligned with concrete syntax of a corresponding DSL.

4. Results

4.1. Metamodel construction

We conducted an extensive literature review to identify common elements, concepts and establish the classification (see Table 1).

This analysis supports the conceptual representation of a technology roadmap as a layered structure with a hierarchy of elements and attributes, connected with causal links and composing a directed graph. This view is well aligned with the architectural framework of Phaal and Muller (2009). Therefore, we took this structure as a foundation of our metamodel.

The identified concepts represent the basis of the proposed MB-TRM (see Fig. 4). The roadmap informational structure is a directed graph consisting of elements and links. Elements of a roadmap exist in the three-dimensional space defined by time, level, and scenario dimensions. Elements may be classified (modeled as "Element references to Element Class"); classes can be organized into hierarchies to form taxonomies (each Element Class can be a parent of other Element Class). Each Element can have several attributes (Attribute) that take quantitative or qualitative values (Attribute Value). In some applications, a corresponding element class defines the set of attributes that Element can exhibit.

We believe that the majority of document-based TRM can be derived from this structure. To realize the full potential of the MB-TRM approach, we added functions and models. When the set of quantitative or qualitative attributes is not sufficient to characterize elements, we may employ models, e.g., structural, parametric, or stochastic. For their part, functions operate on the entire directed graph, execute models, perform calculations of aggregated metrics, and various checks. Functions and models are the exclusive features of MB-TRM and cannot be derived from document-based roadmaps.

The geometric interpretation of the metamodel is shown in Fig. 5 and 6. Points represent Elements, i.e., markets, systems, and technologies. Horizontal causal links represent the evolution of elements. Cross-level links model interdependencies.

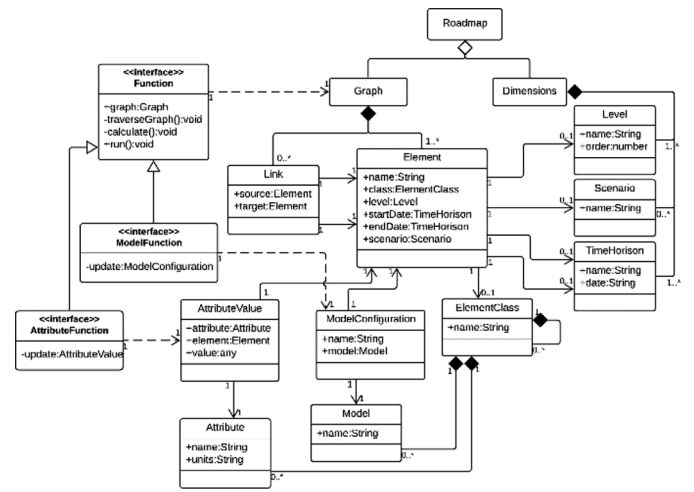


Fig. 4. Metamodel for MB-TRM

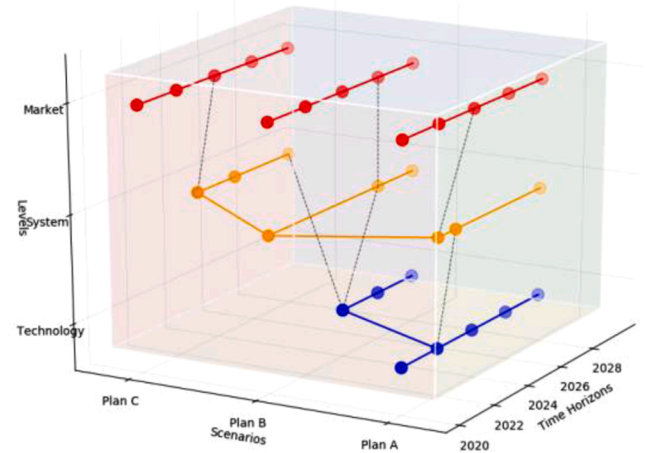


Fig. 5. Geometric interpretation

Depending on the organization, the graph's topology illustrated by Fig. 5 and 6 may be fairly complex. Nodes at each level represent versions/configurations of just one system, technology, or market niche for

Table 1
Document-based technology roadmap commonalities

Reference	Context	Levels	Elements	Attributes	Links
(Albers et al., 2017)	Strategic roadmap in automotive domain	Market, Product, Technology, Resources	Skill, capability, competency, need, product, technology	-	evolves to, satisfies need, requires resource
(Albright and Kappel, 2003)	Product and technology roadmap in telecom. domain	Market, Product, Technology, Risk	Product/market driver, market trend, market segment, product, technology, capability, event, risk	Market growth, market share, experience curve, priority, price, risk consequence, competitive position	requires, targets market niche, evolves to
(Scalice et al., 2015)	TRM for modular products	Product, market, module	Product, platform, market, technology, module	Differentiation feature, lifecycle stage, lifestyle, benefits, driver importance weight, ranking	-
(Toro-Jarrín et al., 2016)	Business Model Canvas and TRM integrated	Market, product, technology	Political strategy, macroeconomic forces, key trends, market and industry forces, new entrants, substitutes, technology, regulatory, cost trends	Economy growth, employment rate, market volume, features, product size	-
(Al-Ali and Phaal, 2019)	TRM for digital transformation	Digital opportunity, digital initiative, resources	Disruptive/Radical opportunities, internal efficiencies, MVP, legacy system, resources, digital technology	-	Replace, interface, migrate
(Zhang et al., 2016)	TRM for technical intelligence in photovoltaic industry	Material, method, device	Problem, solution, component, implication	Efficiency, stability, losses, etc.	Solve, relate, improve, upgrade

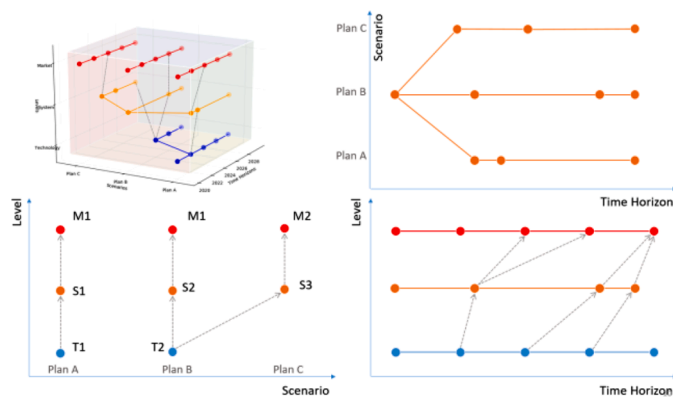


Fig. 6. Geometric interpretation – cross-sections

different moments of time multiplied by the number of alternative realities under consideration (so, three scenarios for four moments of time produce twelve distinct states of the market). Each complex system is composed of hundreds of components and depends on a number of technologies. Technological giants and agencies typically manage tens of complex systems or products/services in the different stages of a life cycle, which are also connected either directly or through the use of the same pool of components/technologies. Connection types are numerous (requires, depends on, excludes, enhances, enables, integrates into, triggers, targets, etc). Moreover, so far, we were mainly talking about roadmap as a model, but each node (or *Element*) has a model itself (ranging from parametric model to a digital mockup). If we go further, such a system would naturally integrate with project management software. In this view, companies' strategic targets can be hierarchically decomposed to project's goals or even to individual tasks. One may argue on the last point that this is not what strategic tools currently doing and whether they should do it at all? However, our main idea here is to illustrate what this type of systems potentially will be able to do. The more data across an enterprise will be collected and aggregated, the more informed strategic decisions are.

4.2. Metamodel validation

To validate the metamodel, we additionally selected three roadmaps (see [table 2](#)) to check whether these roadmaps can be modeled based on the metamodel. We created an abstract and concrete syntax of a DSL and user data layer, precisely representing targeted roadmaps for each of them.

The instantiation of a metamodel for the NASA roadmap is shown in [fig. 7](#). *Technology Area*, *Technology Family*, *Technology Class*, *Technology*, and *Capability* are instances of *Element Class*. They represent NASA's technology taxonomy. *Technology* and *Capability* define the set of attributes that concrete elements of a roadmap – *Technology Current*,

Table 2
Roadmaps selected for validation

Reference	Context	Structural challenge
(2015 NASA Technology Roadmaps)	Technology roadmap of a space agency. Horizon of planning – 30 years.	Relatively complex taxonomy of elements. An enormous amount of data – more than 2000 pages.
(Kim et al., 2018)	UX-centered product, design, and technology roadmap of a single-product startup. Horizon of planning – 3 years.	Simple structure, but data represent subtle and qualitative concepts – user needs and experiences.
(Geum et al., 2014)	Emerging car-sharing business, heavily dependent on the turbulent external environment. Horizon of planning – 30 months.	Alternative realities (scenarios). The abundance of KPIs. A dynamic model of KPIs complements a roadmap.

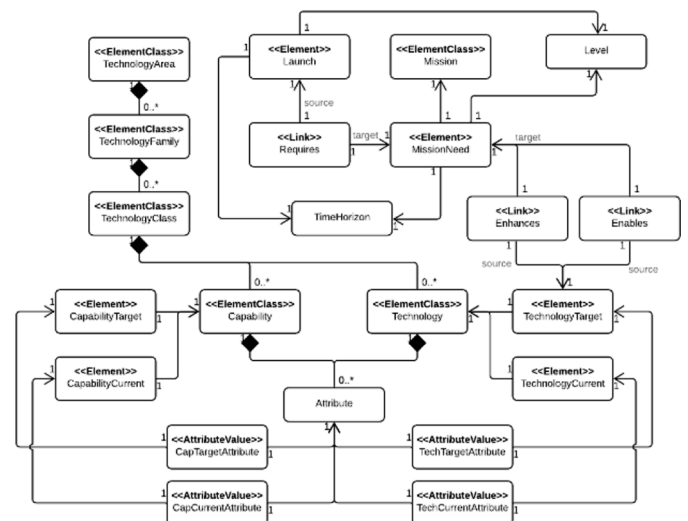


Fig. 7. Instantiation of the metamodel for NASA TRM

Technology Target, *Capability Current*, and *Capability Target* – can take. Another specificity of NASA roadmap is that these elements do not refer to the time dimension directly (note cardinality 0..1 between *Element* and *Time Horizon* in the metamodel). Instead, they are connected with multiple *Mission Needs* by *Enables* and *Enhances* links. This is how different due dates for the same performance target are modeled.

Concrete syntax created for this use case is given in Fig. 8–10. For obvious reasons, we modeled only a small portion of the NASA roadmap. However, we can safely generalize our results on the entire document, thanks to the NASA roadmap’s formal structure.

The 'code completion' (e.g., Design, Product and Technology levels on Fig. 12) and the 'intention actions' (e.g., 'Technology Parameter' on Fig. 9) dialogs are shown on these and the following figures. This is done to illustrate that in MPS we do not write code to be parsed and compiled but directly manipulate AST, which is the major premise for the validation: strict compliance of every written expression to defined abstract syntax is ensured.

The instantiation of the metamodel for the design roadmap is shown on Fig. 11. The structure consists of 9 *Element* instances and 3 *Link* instances. The meaning of each *Element* is characterized by a textual description.

Some of the classes are not connected (Fig. 11) to others by association links. It does not mean that they are unrelated to other parts of the model. The model of level M1 inherits the relationships defined by the metamodel (level M2). These general relationships may be additionally

```
Technology Area 1 Launch Propulsion Systems
Technology Family 1.1 Solid Rocket Propulsion Systems
Technology Class 1.1.1 Propellants
Technology Class 1.1.2 Case Material
Technology Class 1.1.3 Nozzle System
Technology Class 1.1.4 Hybrid Rocket
Technology Class 1.1.5 Fundamental Solid Propulsion Technologies
Technology Class 1.1.6 Integrated Solid Motor Systems
Technology Class 1.1.7 Liner and Insulation
Technology Family 1.2 Liquid Rocket Propulsion Systems
...
Technology Family 1.3 Air-Breathing Propulsion Systems
...
Technology Family 1.4 Ancillary Propulsion Systems
...
Technology Family 1.5 Unconventional and Other Propulsion Systems
...
Technology Family 1.6 Balloon Systems
...
Technology Area 2 In-Space Propulsion Technologies
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Fig. 8. Concrete syntax for NASA TRM – classifier

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Insulating/Ablative Sprayable Liner is a Technology
with attributes:
  Current case material weigh units: %
Technology Description:
  This is a liner capable of insulative properties similar to fiber-filled nitrile butadiene rub
Technology Challenge: Producing an acceptable sprayable liner is a technical challenge.
Technology Development Dependent Upon Basic Research: false
Technology Class: 1.1.7 Liner and Insulation
Classifier: 1.1.7.2
Technology State of the Art:
  This technology is under development in Space Launch System (SLS) Booster project. Current issues
TRL: 5
define attribute values:
  [redacted]
Tech Technology Parameter (AttributeValue in M2_AbstractLanguage) /ation thickness
TRL: 9
define attribute values:
  Current case material weigh 25 %
A sprayable liner is a Capability
with attributes:
  Current case material weigh units: %
Capability Description:
  The sprayable liner can maintain the solid rocket booster motor case internal temperature below

```

Fig. 9. Concrete syntax for NASA TRM – technology editor

```

DRMS Asteroid Redirect - Robotic spacecraft is a Mission

create Launch Launch_DRMS
( start date: 2018 level: Human exploration and Operations scenario: <no scenario> )

create Need for DRMS Asteroid Redirect - Robotic spacecraft Need_DRMS
( epoch: 2015 level: Human exploration and Operations scenario: <no scenario> )

1.1.7.1 Performance Goal enables Need_DRMS
1.1.7.2 Performance Goal enables Need_DRMS

Press [Enter] to define enhancer links

Launch_DRMS requires Need_DRMS

```

Fig. 10. Concrete syntax for NASA TRM – roadmap editor

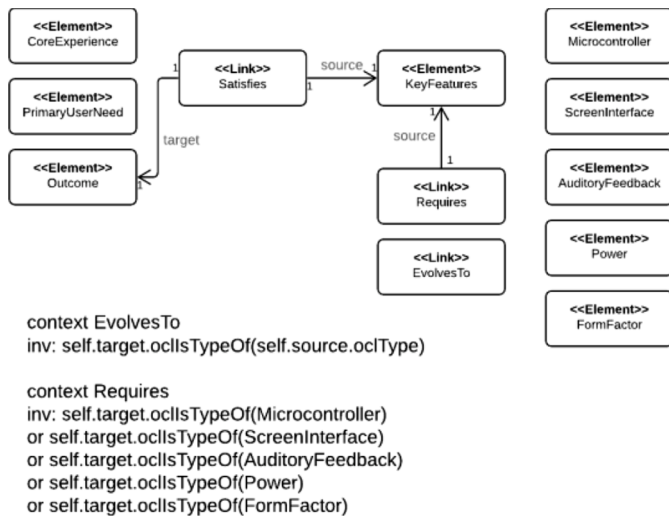


Fig. 11. Instantiation of the metamodel for Design roadmap

specified on the level M1 with OCL-expressions. For instance, a target and a source of the link *Evolves to* should be of the same concept, as it signifies the evolution of needs/products/technologies. Furthermore, link *Requires* connects *Key Features* and any *Element* that models technology. At the same time, link *Satisfies* can go strictly from *Key Features* to *Outcomes*. Fragments of a concrete syntax for the design roadmap are given in Figs. 12–13.

Finally, the instantiation of a metamodel for the emerging car-sharing business (Geum et al., 2014) is shown in fig. 14. This is an example of a roadmap modeled in a three-dimensional space. Casual links connect *Business Drivers*, *Products*, *Services*, and *Technologies* either within the same scenario or produce scenario branching by connecting two *Elements* placed in distinct scenario planes. In the original paper of Geum et al. (2014), such a structure was used to launch a dynamic

```

Core experiences:
create Core Experience Core experience 1
( start date: 2012 level: Design Roadmap scenario: <no scenario> )
description: Make learning about diabetes fun through game play
create Core Experience Core experience 2
( start date: 2013 level: Design Roadmap scenario: <no scenario> )
description: Learn about diabetes through a combo of free and guided play
create Core Experience Core experience 3
( start date: 2015 level: Design Roadmap scenario: <no scenario> )
description: Learn about c
  * Design Roadmap ^levels (M1_2_DesignRoadmap.null)
  * Product Generation ^levels (M1_2_DesignRoadmap.null)
  * Technology Development ^levels (M1_2_DesignRoadmap.null)
Primary User Needs:
create Primary User Need User need 1
( start date: 2012 level: Design Roadmap scenario: <no scenario> )
description: Learning about diabetes should be fun and easy
create Primary User Need User need 2

```

Fig. 12. Concrete syntax for Design roadmap – Elements

```

Form-factor:
create Form Factor A Bear shell-1 (Teddy Bear)
( start date: 2012 level: Technology Development scenario: <no scenario> )
description:
create Form Factor A Bear shell-2 (Teddy Bear)
( start date: 2013 level: Technology Development scenario: <no scenario> )
description: easy to clean Asthma-friendly fabric

Links:
Generation 1 satisfies Outcome 1
Generation 1 requires Arduino
Generation 2 satisfies Outcome 2
Generation 3 requires Co
  * Generation 1 ^key_features (M1_2_DesignRoadmap.null)
  * Generation 2 ^key_features (M1_2_DesignRoadmap.null)
  * Generation 3 ^key_features (M1_2_DesignRoadmap.null)
Generation 2 requires Co
Generation 1 evolves to
Generation 2 evolves to Generation 3

```

Fig. 13. Concrete syntax for Design roadmap – Links

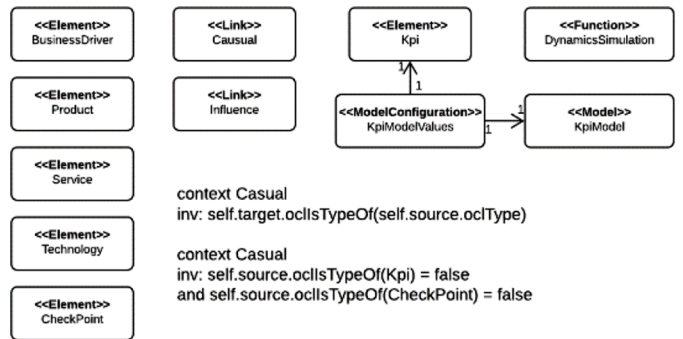


Fig. 14. Instantiation of the metamodel for emerging car-sharing business roadmap

simulation to calculate business KPI's for pessimistic/neutral/optimistic scenarios. The interdependencies between KPI's and elements of a roadmap were defined by 'causal and loop diagram.' Our metamodel is compatible with such a problem setting: a mutual influence between roadmap elements and KPI's is established by the *Influence* link, the transfer function of each KPI – in the *KpiModel*, and initial values – in the *Kpi Model Values*. The simulation is executed by *Dynamic Simulation* function that traverses graphs and iteratively updates *KpiModelValues*.

```

Add Product Delete
create Product Development of environmental-friendly car
( start date: Jan-2010 end date: Mar-2011 level: Product scenario: Neutral )
create Product Automatic payment function
( start date: Jul-2011 end date: Jan-2012 level: Product scenario: Neutral )
create Product Car for ergonomics
( start date: Jul-2010 end date: May-2011 level: Product scenario: Neutral )
create Product Car for multiple customers
( start date: Jan-2010 end date: Jul-2010 level: Product scenario: Neutral )
Add Service Delete
create Service e-Tracking service
( start date: Jul-2011 end date: Apr-2012 level: Service scenario: Neutral )
create Service Automatic return service
( start date: May-2011 end date: Nov-2011 level: Service scenario: Neutral )
create Service Public Transp. notification

```

Fig. 15. Elements of a roadmap definition

The examples of a concrete syntax for this use case are in –Figure 15, 16 and 17.

5. Discussion

During the validation, we were able to assert all three of our statements (S1-S3).

S1: we successfully created three DSLs that could precisely represent the information contained in the three publicly available roadmaps. By doing so, the internal consistency of our metamodel is validated. Second, we demonstrated the ability of the proposed abstract syntax to describe real roadmaps correctly.

S2: we deliberately selected extremely different roadmaps: the roadmap of the space agency versus the roadmaps of the startups; the descriptive roadmaps used for external and internal communication versus the executable roadmap used for simulation. By illustrating different case studies, we aimed to show that our metamodel applies to a wide range of contexts.

S3: our metamodel complies with MB-TRM principles and introduces concepts that extend conventional technology roadmaps: element models and graph traversal functions.

However, several research limitations can be identified and discussed. Being not a natural phenomenon but a kind of collective pragmatic knowledge, TRM's notion can be subject to different interpretations. Therefore, we have opted for the analysis of the scientific literature instead of seeking expert opinion. Based on that, we are convinced that our metamodel corresponds to what most academics mean by this notion. At the same time, we are aware of the possibility of alternative views.

There is a large variety of TRMs, which additionally integrate other methods of strategic management. We do not claim to have presented an all-encompassing metamodel. However, we do believe that the proposed meta model is applicable to a broad set of TRMs.

The other problem of our methodology is selection and confirmation biases: a set of roadmaps used for analysis was composed by the authors of this contribution. Therefore, it may be interesting to see the alternative variants of the metamodel and, consequently, the consensus version.

Finally, focusing on the informational properties, we have entirely ignored the part of the architecture responsible for the visualization of all this information. It is a notable omission as roadmapping is essentially a visual approach that needs to be addressed in the future.

6. Conclusions and future work

In this paper, we propose a metamodel for the MB-TRM, a new concept in the field of technology management and strategic planning. We demonstrate that our metamodel can be instantiated to three roadmaps, sufficiently different in terms of the context and primary objectives. The latter indicates that our meta model can be used in practice as a generalization for the development of new software tools.

Moreover, our proposal helps to bridge the gap between the technology roadmapping domain and the MDE domain. Furthermore, we propose an original method to validate a metamodel using a modern language workbench. This can be considered as a methodological contribution and might be beneficial for researchers dealing with metamodeling in the field of systems engineering and design science.

In contrast to computer-based approaches, which focus mainly on data gathering for the technology forecasting stage, MB-TRM targets the process of roadmapping itself. The main idea behind MB-TRM is motivated by the fact that in big organizations, the scope of the roadmap is vast and complex (the reader can take an open-source NASA roadmap as a reference to illustrate the degree of complexity). Even though the resulting layered chart may and should be relatively simple, the amount of underlying information that supports these strategic decisions can be potentially unmanageable. During the roadmapping session, it is usually

Fig. 16. Causal links definition

Fig. 17. Dynamic model set up

necessary to zoom in/out, dive into layers, and reflect on different cross-sections of the multidimensional domain model. Such operations facilitate checks of adjacent departments' roadmaps and align plans with design teams, suppliers, and overall company strategy. The required flexibility, interactivity, and automation, absolutely unreachable when static documents and reports are used, can be, however, achieved within the model-based paradigm.

To illustrate the promises and challenges of the approach, in our next work, we intend to apply the MB-TRM to help one company from the automotive industry in planning their new product line. We aim to instantiate the proposed meta model, and implement an MB-TRM tool to the degree of the usable interactive application. User-centered design problematics is especially challenging for model-based approach (compared to, for instance, the aerospace domain), as the user experience-related concepts are subjective, ambiguous, and hardly quantifiable.

CRedit authorship contribution statement

Ilya Yuskevich: Methodology, Software, Validation, Investigation, Formal analysis, Writing – original draft. **Andreas Makoto Hein:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Kahina Amokrane-Ferka:** Conceptualization, Writing – review & editing, Project administration. **Abdelkrim Doufene:** Conceptualization, Methodology, Writing – review & editing, Funding acquisition. **Marija Jankovic:** Conceptualization, Writing – review & editing, Supervision.

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Ilya Yuskevich is a doctoral student at CentraleSupélec, University Paris-Saclay, and at IRT SystemX (France). He holds a Specialist Degree in Radio electronics from Bauman University (Russia) and a Master's degree in Space Systems Engineering from Skoltech (Russia). His career started in the field of microwave devices and antenna arrays development for marine and space applications. His professional interests later shifted to the conceptual design of satellite systems. Currently, he is working on the design of methods and software tools for new product development. His research interests encompass technology planning and road mapping, user-centered design, and model-based systems engineering.

Dr. Andreas Hein is a system architect at IRT-SystemX. He was formerly an assistant professor at the Industrial Engineering Lab at CentraleSupélec – Université Paris-Saclay in the area of systems engineering. He obtained his Bachelor's and Master's degree in aerospace engineering from the Technical University of Munich and conducted his PhD research on heritage technologies in space programs at the same university and at the Massachusetts Institute of Technology. Andreas has published over 60 articles in peer-

reviewed international journals and conferences. He is the recipient of the Exemplary Systems Engineering Doctoral Dissertation Award and the Willy Messerschmitt Award.

Dr. Kahina Amokrane-Ferka is a senior research engineer in human-machine interaction. She is currently working as a system architect at IRT SystemX, France. Her education includes a Computer Science degree from UMMTO University, Algeria, in 2005, and then a Master's research degree from Evry Val d'Essonne University, France, in 2006. She received a Ph.D. degree in Information Technologies and Systems from the University of Technology of Compiègne in 2010 and then worked as an academic researcher for four years. Her research interests include knowledge management, human factors, simulation, intelligent system, system engineering, system modeling, serious gaming, and human systems integration. She has also taught courses in Algorithmic and system modeling at the UTC and CentraleSupélec University.

Dr. Abdelkrim Doufene is the Director of Strategy and Programs at the Institute for Technological Research (IRT) SystemX. His main mission is to maintain and update the institute's vision and research roadmaps. He contributes to consolidate the position of SystemX in its ecosystem and coordinates all the project lifecycle activities, and the asset

management strategy. Abdelkrim has been a Research Scientist at Massachusetts Institute of Technology and a research engineer at Renault. He has taught in many universities and schools. Abdelkrim holds an engineering degree in computer science from ESI, Algeria, and a master degree in decision and information systems from Université Paris 1 Panthéon-Sorbonne, ESSEC and CNAM. He obtained a Ph.D. from École polytechnique France.

Dr. Marija Jankovic is a Professor at Ecole Centrale Paris (ECP), now CentraleSupélec, University Paris-Saclay. The research focus of Dr. Jankovic is on supporting early design stages, in particular in System Architecture design integrating innovation and uncertainty management. The aim is to support design team and their decision-making process in the systematic exploration of the design space as well as supporting decision making with regard to uncertainties inherent to early design. The approaches developed are in line with Complexity science and the relational theory of complexity management, underlining relationships, and exploring design synthesis methods with regard to a large number of parameters and system performance indicators. Dr. Jankovic is interested in the decision making in design, DSM (Design Structure Matrix) and other relational modeling and simulation of complexity, System Architecture design, Uncertainty modeling and management, Design Synthesis.